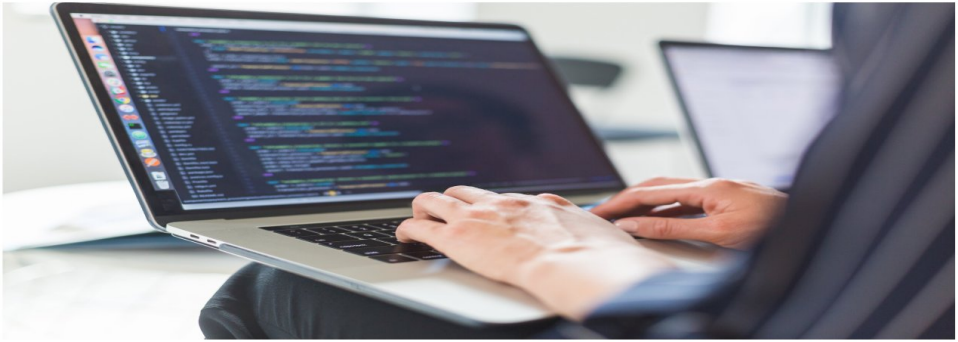


ETX1100/ETX5900 Business Statistics
Week 7 Part 1: Correlation and Regression Analysis
Part 2: Time Series Analysis & Forecasting
Workshop Lead: Alyssa Hu

Updates and Reminders

- Next week, our workshop will be dedicated to summary and revision including practice problems
- The teaching team will indicate pre-exam consultation schedule on Moodle
- The mock exam and past exams are available under Module 8
- There is a Chatbot (All Day TA) and you can practice unit-related questions using the Intelligent Quiz tab

Week 7: Learning Outcomes

Multiple Linear Regression:

- Understand the concept of omitted variable bias
- Compare and evaluate slope coefficients in a simple and multiple linear regression
- Perform hypothesis tests to check for a significant relationship between independent variables and the dependent variable.
- Understand measures by which we judge the fit of a regression model
- Predict with multiple regression model

Time Series:

- Identifying time series components: trend, seasonal, cycle and irregular
- Modeling the trend and seasonality components.
- Understand and apply MAD and MSFE
- Forecasting out of sample values using the time series regression model to aid decision-making in business

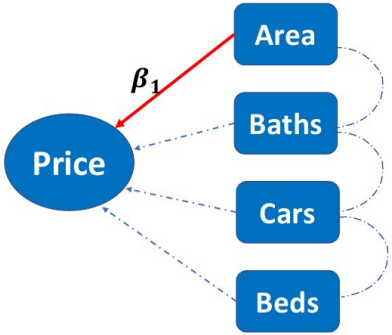
Textbook references:
Berenson et al *Basic Business Statistics 5th edition,*

Simple Linear Regression Pitfalls

So far, we've only included a SINGLE independent variable.

Omitting other relevant variables from the regression model may result in biased estimates of the regression coefficient (omitted variable bias).

e.g. Larger properties (in land area) tend to have larger dwelling size and therefore, more beds, baths and cars.



Simple Linear Regression Pitfalls

We *could* estimate several simple linear regressions:

	Intercept	Slope
Area	539.357	1.093
Beds	261.501	264.855
Baths	661.001	288.180
Cars	845.000	200.464

These results seemingly give you area, beds, baths and cars premiums (i.e. how these variables increase the price).

But they are misleading because of the interconnected relations between variables!

Multiple Linear Regression (MLR)

The dependent variable Y may depend on several independent variables…

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon_i$$

Independent variables $X_1, X_2, \ldots, X_k$ are related to Y. But some of the independent variables may also be related.

We can generate a multiple regression output in Excel. But be careful:

Our interpretation for $\beta_1, \beta_2, \ldots, \beta_k$ is different from simple linear regression!

Multiple Linear Regression (MLR)

$$\hat{Y}_i = b_0 + b_1 X_1 + b_2 X_2 + \cdots + b_k X_k$$

b_0: The predicted value of Y when all X variables are zero
Not meaningful when the range value of any X variables does not include 0

b_1: The change in predicted Y when X_1 value increases by one unit, holding other X variables fixed.

...

b_k: The change in predicted Y when X_k value increases by one unit, holding other X variables fixed.

Important: The slope estimates in MLR are **marginal effects**, i.e. the effect of a single X input holding other X variables fixed.
(In SLR, the slope tells you the *total effect* of X on Y because there were no other variables to

Guided Example: Real Estate Data Multiple Linear Regression

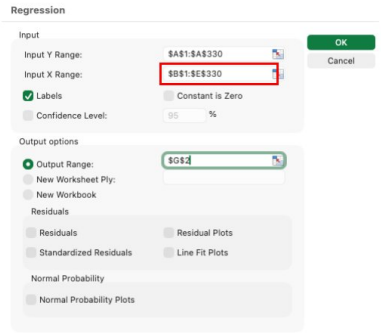
Using the Real Estate data, answer the following questions:

- Generate a multiple linear regression output using Microsoft Excel
- Interpret the slope coefficients for **Beds and Area**
- Compare the simple and multiple linear coefficients from your outputs
- Test whether the number of bedrooms has a positive impact on Price
- Select the best model using hypothesis testing
- Interpret the coefficient of multiple determination
- Test for the significance of the overall multiple linear regression model
- Predict the price of a property that is 720sqm in area, with 4 bedrooms, 2 bathrooms & 2 car spaces

Guided Example: Generate a multiple linear regression

Input X range now covers **multiple columns.**

Note, all X's must be stored in **consecutive columns** in your spreadsheet.



Guided Example: Multiple Regression Output

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.777331023					
R Square	0.60424352					
Adjusted R Square	0.599357637					
Standard Error	265.1623433					
Observations	329					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	4	34781849.65	8695462.413	123.671317	6.00458E-64	
Residual	324	22780786.12	70311.06828			
Total	328	57562635.78				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	86.84134712	61.99609571	1.400755098	0.162244312	-35.12436376	208.807058
Beds	47.42619189	22.86632063	2.074063102	0.038862943	2.440987277	92.41139651
Baths	216.5331552	21.40811948	10.11453413	4.42491E-21	174.4166884	258.6496221
Cars	14.54447501	22.7739893	0.638644149	0.523505634	-30.25908501	59.34803502
Area	0.83126655	0.075739751	10.97530081	4.85461E-24	0.682262771	0.980270329

Guided Example: Multiple Regression Output

$$\widehat{Price}_i = 86.841 + 47.426Beds_i + 216.533Bath_i + 14.544Cars_i + 0.831Area_i$$

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.777331023					
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Guided Example: Interpretation of the slope coefficients

$$\widehat{Price}_i = \mathbf{86.841} + \mathbf{47.426}Beds_i + \mathbf{216.533}Bath_i + \mathbf{14.544}Cars_i + \mathbf{0.831}Area_i$$

Interpret b_1 (slope coefficient for beds) and b_4 (slope coefficient for area) .

Guided Example: Interpretation of the slope coefficients

$$\widehat{Price}_i = \mathbf{86.841} + \mathbf{47.426}Beds_i + \mathbf{216.533}Bath_i + \mathbf{14.544}Cars_i + \mathbf{0.831}Area_i$$

Interpret b_1 (slope coefficient for beds) and b_4 (slope coefficient for area) .

The estimated premium (increase in price) of an **additional bedroom,** given the baths, cars and area features remain fixed, is $47,426 on average.

- If you are comparing two properties with exactly the same land size, car spaces, and bathrooms, a property with 4 bedrooms is expected to cost $47,426 more than a property with 3 bedrooms.

The estimated premium (increase in price) of an **additional square metre** of land area, given the beds, baths and cars features remain fixed, is $831 on average.

- If you are comparing two properties with exactly the same number of beds, baths, and car spaces, the property that is 780 sqm is expected to cost $ 8,310 more than the property that is 770 sqm

Guided Example: Compare SLR and MLR

	Simple		Multiple	
	Intercept	Slope	Intercept	Slope
Area	539.357	1.093	86.841	0.831
Beds	261.501	264.855		47.426
Baths	661.001	288.180		216.533
Cars	845.000	200.464		14.544

- **Simple linear regression overestimates all four premiums**
- Bathrooms remain the premium feature with >$200k premium for an additional bathroom, given other features remain fixed
- Car spaces in the simple linear regression yield a large premium (>$200k), but became negligible after other factors are accounted for (<$15k)

Guided Example: Testing for Significance of the Slope

Similar to what we did with simple linear regression:

H_0 says there is no linear relationship between X_k and Y

H_1 is where we can test for a linear relationship

H_0: $\beta_k = 0$ H_1: $\beta_k \neq 0$ Test if there is a linear relationship between X_k and Y (two-tailed test)	H_0: $\beta_k = 0$ (or: H_0: $\beta_k \geq 0$) H_1: $\beta_k < 0$ Test if X_k has a negative linear effect on Y (left-tailed test)	H_0: $\beta_k = 0$ (or H_0: $\beta_k \leq 0$) H_1: $\beta_k > 0$ Test if X_k has a positive linear effect on Y (right-tailed test)

Guided Example: Testing for Significance of the Slope

We want to test whether the number of bedrooms has a positive impact on Price

		Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
β_0	Intercept	86.84134712	61.99609571	1.400755098	0.162244312	-35.12436376	208.807058
β_1	Beds	47.42619189	22.86632063	2.074063102	0.038862943	2.440987277	92.41139651
β_2	Baths	216.5331552	21.40811948	10.11453413	4.42491E-21	174.4166884	258.6496221
β_3	Cars	14.54447501	22.7739893	0.638644149	0.523505634	-30.25908501	59.34803502
β_4	Area	0.83126655	0.075739751	10.97530081	4.85461E-24	0.682262771	0.980270329

p−value for a two−tailed test

Be careful! Excel reports two-tailed p-values. What type of test are we doing?

Guided Example: Testing for Significance of the Slope

We want to test whether the number of bedrooms has a positive impact on Price

		Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
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β_3	Cars	14.54447501	22.7739893	0.638644149	0.523505634	-30.25908501	59.34803502
β_4	Area	0.83126655	0.075739751	10.97530081	4.85461E-24	0.682262771	0.980270329

$H_0: \beta_1 = 0$ (No relationship between Price and Beds)
$H_1: \beta_1 > 0$ (Beds has a positive impact on Price)

α = 0.05, t-statistic is 2.074

p-value $= 0.03886/2$ = 0.01943 < 0.05.

Therefore, we reject H_0

There is sufficient evidence at the 5% level of significance that the number of bedrooms has a positive impact on property prices

Guided Example: Select the Best Model with Hypothesis Testing

- Hypothesis testing for the regression coefficients can be used to select the best model, as not all independent variables are necessary.
- After fitting a regression with all potential independent variables, remove statistically insignificant variables from the model.

$H_0: \beta_k = 0$ (no relationship between the dependent variable and the independent variable X_k)
$H_1: \beta_k \neq 0$ (the independent variable X_k, affects the dependent variable)





Reject $H_0: \beta_k = 0$ when p-value $< \alpha$
- Variable X_k affects the dependent variable
- **Keep** X_k in the model

Do not reject $H_0: \beta_k = 0$ when p-value $> \alpha$
- No relationship between dependent variable & X_k
- Consider to **remove** X_k from the model

Guided Example: Model Reduction

Since Cars has no significant impact on Price, we can consider removing it from the model:

- Create a new sheet of data
- Remove Cars by right clicking on column D and select Delete
- Estimate the model again with the remaining inputs
- Excel requires all explanatory variables (X's) to be in consecutive columns

Guided Example: Model Reduction

Reduced model:

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.777010504					
R Square	0.603745323					
Adjusted R Square	0.600087588					
Standard Error	264.9206769					
Observations	329					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	3	34753172.13	11584390.71	165.0598646	5.26039E-65	
Residual	325	22809463.64	70182.96505			
Total	328	57562635.78				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	94.29534357	60.83202476	1.550093786	0.12209221	-25.37889511	213.9695822
Beds	51.36298726	21.99969527	2.334713579	0.020166972	8.083205065	94.64276946
Baths	217.7596943	21.30236672	10.22232399	1.87828E-21	175.8517596	259.667629
Area	0.83592956	0.075318289	11.09862655	1.74896E-24	0.687756637	0.984102482

Coefficients of the remaining factors did not change drastically

Guided Example: Coefficient of Multiple Determination (R^2)

Let's proceed with this reduced model:

Proportion of the variation in Y that is explained by the set of independent variables?

Guided Example: Coefficient of Multiple Determination (R^2)

Let's proceed with this reduced model:

Proportion of the variation in Y that is explained by the set of independent variables?

R^2 = 0.6037

- About 60% of the variation in price is explained by the reduced model (with variables beds, baths and area).
- About 40% of the variation in price remains unexplained.

Note: R^2 is higher for the multiple regression model than for each of the simple linear regression models. Every time a variable is added, R^2 accounts for more of the variability in the dependent (response) variable.

Test for the Significance of the Overall Multiple Regression Model

We can ask if there is an overall fit, i.e. Is the linear relationship between the response variable *Y* and the explanatory variables $X_1, X_2, \ldots, X_k$ valid?

Or

Is there a significant relationship between the dependent variable and the entire set of independent variables?

The hypotheses:

H_0: $\beta_1 = \beta_2 = \ldots = \beta_k = 0$

H_1: At least one $\beta_i \neq 0$

No linear relationship between the dependent variable and the independent variables

Linear relationship between the dependent variable and the independent variables

- What does this mean?

Test for the Significance of the Overall Multiple Regression Model

• If H_0 is rejected,
this means that the explanatory variables as a group provides at least some power to explain the variation in the response variable Y.

• If H_0 is not rejected
this means that this set of explanatory variables has no power to explain the variation in the response variable Y.

We use the *F*-test to test the significance of the overall multiple regression model. The *F*-test is a formal procedure for testing whether the explained variation is large compared to the unexplained variation.

Guided Example: Hypothesis Test for Overall Fit (F-test)

At the 5% level of significance, test if there is a significant linear relationship between Price and the three independent variables of beds, baths, and area.

Step 1:

H_0:

H_1:

Step 2:

α = 0.05

F statistic = ?

Step 3:

P-value = ?

Step 4:

Decision:

Reject H_0 if p-value < α

Fail to reject H_0 otherwise

Step 5:

Conclusion

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.777010504
R Square	0.603745323
Adjusted R Square	0.600087588
Standard Error	264.9206769
Observations	329

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	34753172.13	11584390.71	165.0598646	5.26039E-65
Residual	325	22809463.64	70182.96505		
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	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
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Guided Example: Hypothesis Test for Overall Fit (F-test)

At the 5% level of significance, test if there is a significant linear relationship between Price and the three independent variables of beds, baths, and area.

Step 1:

H_0: $\beta_1 = \beta_2 = \beta_3 = 0$

H_1: *at least one* $\beta_i \neq 0$

(where $i = 1,2,3$)

Step 2: α = 0.05, F statistic: 165.060

Step 3:

p-value (Significance F) = 5.260×10^{-65}

Step 4: Reject H_0 if p-value < α

$5.260 \times 10^{-65} < 0.05$, reject H_0

Step 5: There is sufficient evidence at 5% significance level to conclude that the linear relationship between Price and the combination of beds, bath, and area is valid (i.e., at least one of the independent variables is related to Price)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.777010504
R Square	0.603745323
Adjusted R Square	0.600087588
Standard Error	264.9206769
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ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
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Guided Example: Prediction using Regression Model

$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \ldots + \hat{\beta}_k X_k$ OR $\hat{Y}_i = b_0 + b_1 X_1 + b_2 X_2 + \ldots + b_k X_k$

- $\hat{Y}_i$ allows us to use information on $X's$ to predict *Y*.
- Plug in the value of *X* for each independent variable, multiply it with the corresponding estimated coefficient, then take the total sum to get $\hat{Y}_i$!

Best practices for prediction:

- Make a prediction for the dependent variable only after **confirming the significance** of the regression coefficients through hypothesis testing!
- Make prediction for values of $X's$ that are **inside the range of $X's$ values** in the original sample

Guided Example: Real Estate Data - Prediction

- From the regression model, we can predict the price of a property
- **Consider a 720sqm property, with 4 bedrooms, 2 bathrooms & 2 car spaces**
- We can use the =SUMPRODUCT() function

Guided Example: Real Estate Data - Prediction

- From the regression model, we can predict the price of a property
- **Consider a 720sqm property, with 4 bedrooms, 2 bathrooms & 2 car spaces**
- We can use the =SUMPRODUCT() function to get approximately $1,337,136

Estimates of $\beta's$ from regression

	K	L	M	N	O
	Prediction for a 720sqm property, 4 bedrooms, 2 bathrooms, 2 carspaces				
		Coefficients	Input		
	Intercept	94.29534357	1		
	Beds	51.36298726	4		
	Baths	217.7596943	2		
	Area	0.83592956	720		
	Prediction	=SUMPRODUCT(L4:L7,M4:M7)			

The input $X's$ for the prediction

Task 1: Climate change

Climate change and a country's carbon footprint are contentious issues faced by many governments. Limiting or reducing carbon dioxide (CO_2) emissions is seen as a priority by climate change activists. Here we attempt to predict CO_2 emissions (million metric tonnes) for 14 countries based upon their GDP (in US $trillion) and population density (people per square kilometre). Refer to the ***Emissions*** data worksheet.

a. State the estimated multiple regression equation. Define any symbols used. Predict the CO_2 emissions for a country with GDP of $1 trillion and a population density of 50 people per square kilometre. Is this a valid prediction?
b. Interpret the slope coefficient population density in the estimated equation. Is it as expected?
c. State the coefficient of determination and interpret its meaning in the context of the question.
d. At the 5% level of significance, test if there is a significant linear relationship between CO_2 emissions and the two independent variables. (Test for the Significance of the Overall Multiple Regression Model)
e. At the 5% level of significance, whether **each** independent variable makes a contribution to the regression model

Task 1: Climate change

$$\widehat{Emissions} = 151.393 + 204.640GDP - 0.164PopDens$$

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.73305364
R Square	0.537367639
Adjusted R Square	0.453252664
Standard Error	866.5677347
Observations	14

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	9594740.095	4797370	6.38849	0.014414449
Residual	11	8260336.027	750940		
Total	13	17855076.12			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	151.3934513	382.922507	0.39536	0.70013	-691.413304	994.200207
GDP	204.6396651	57.47540943	3.56047	0.00447	78.13714183	331.142188
Population density	-0.16385399	1.550868637	-0.1057	0.91776	-3.57729284	3.24958487

Task 1: Climate change

$\hat{Y} = 151.393 + 204.640X_1 - 0.164X_2$

$\hat{Y}$: estimated CO2 emissions (million metric tonnes)

X_1 : GDP (trillion $)

X_2 : population density (people per square km)

$\hat{Y} = 151.3935 + 204.6397 * 1 - 0.16385*50$

$\hat{Y}$ = 347.84 million metric tonnes

Validity: The prediction is valid because GDP = $1 trillion and population density = 50 people per square kilometre is within the range of the data.

Task 1: Climate change

b. Interpret the slope coefficient population density in the estimated equation. Is it as expected?

b_1 = -0.164

Interpretation: On average, when population density increases by 1 person per kilometre, the CO_2 emissions is estimated to decrease by 0.164 million metric tonnes, holding constant the GDP.

As expected?

It is not as expected because we would expect CO_2 emissions to increase as population density increases. However, our sample indicates otherwise. This can be due to the fact that in the dataset there are some countries with a relatively low population density but very high CO_2 emissions (e.g., Russia and US).

Task 1: Climate change

c. State the coefficient of determination and interpret its meaning in the context of the question.

R^2 = 53.74%

Approximately, 53.74% of the **variation** in CO_2 emissions is explained by the regression model.

The remaining 46.26% of the **variation** in CO_2 emissions is left unexplained by the model.

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Population density	-0.16385399	1.550868637	-0.1057	0.91776	-3.57729284	3.24958487

Task 1: Climate change

d. At the 5% level of significance, test if there is a significant linear relationship between CO_2 emissions and the two independent variables.

Step 1:

Step 2:

Step 3:

Step 4:

Step 5:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.73305364
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Total	13	17855076.12			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	151.3934513	382.922507	0.39536	0.70013	-691.413304	994.200207
GDP	204.6396651	57.47540943	3.56047	0.00447	78.13714183	331.142188
Population density	-0.16385399	1.550868637	-0.1057	0.91776	-3.57729284	3.24958487

Task 1: Climate change

d. At the 5% level of significance, test if there is a significant linear relationship between CO_2 emissions and the two independent variables.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.73305364
R Square	0.537367639
Adjusted R Square	0.453252664
Standard Error	866.5677347
Observations	14

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	9594740.095	4797370	6.38849	0.014414449
Residual	11	8260336.027	750940		
Total	13	17855076.12			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	151.3934513	382.922507	0.39536	0.70013	-691.413304	994.200207
GDP	204.6396651	57.47540943	3.56047	0.00447	78.13714183	331.142188
Population density	-0.16385399	1.550868637	-0.1057	0.91776	-3.57729284	3.24958487

Step 1:

H_0: $\beta_1 = \beta_2 = 0$

H_1: $at\ least\ one\ \beta_i \neq 0$ (where i=1,2)

Step 2: α = 5%, F statistic: 6.3885

Step 3:

P-value (Significance F) = 0.0144

Step 4: Reject H_0 if p-value < α

0.0144 < 0.05, reject H_0

Step 5: There is sufficient evidence at 5% level of significance to conclude that the linear relationship between CO_2 emissions and the combination of GDP and population density is valid (at least one of the independent variables is related to CO_2 emissions)

Task 1: Climate change

e. At the 5% level of significance, test whether each independent variable makes a significant contribution to the regression model.

Hint: Hypotheses will look like

H_0: $\beta_k = 0$ $\quad$ H_1: $\beta_k \neq 0$

Replace k with the variable name (or number)

For *GDP:*

t-statistic = 3.5605

p-value = 0.00447 < 0.05

Reject H_0

There is sufficient evidence at the 5% level of significance that GDP contributes significantly to the explanation of CO_2 emissions.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.73305364					
R Square	0.537367639					
Adjusted R Square	0.453252664					
Standard Error	866.5677347					
Observations	14					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	2	9594740.095	4797370	6.38849	0.014414449	
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Population density	-0.16385399	1.550868637	-0.1057	0.91776	-3.57729284	3.24958487

Task 1: Climate change

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.73305364					
R Square	0.537367639					
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GDP	204.6396651	57.47540943	3.56047	0.00447	78.13714183	331.142188
Population density	-0.16385399	1.550868637	-0.1057	0.91776	-3.57729284	3.249584871

For *Population density:*

t-statistic = -0.1057

p-value = 0.9177 > 0.05

Do not reject H_0

There is insufficient evidence at the 5% level of significance that population density contributes significantly to the explanation of CO_2 emissions

Correlation *NOT* Causation

- Unless the data is obtained in a carefully controlled experiment - not the case here - we can never make definitive statements about causation in regression analysis.

- The reason for this is that we can almost never rule out the possibility that some other variable is causing the variation in *both* of the observed variables.

Example:

- Obtain the data on all the fires in Melbourne for the last ten years.
- The ***number of fire fighters*** at a fire correlates highly with the corresponding ***damage in dollars***.
- Can we conclude that fire fighters cause the damage???

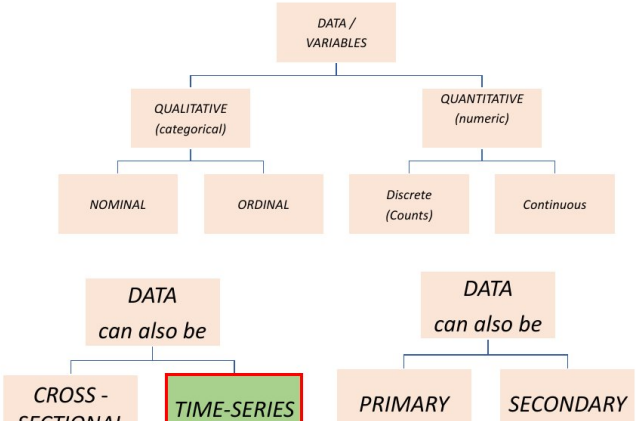
  No, of course not! Both variables result from and are correlated with the overall size of the fire.



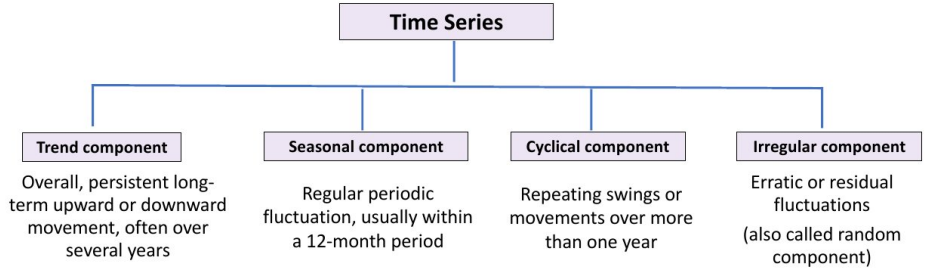


Part II: Time Series Analysis & Forecasting

Data and Variable Classification

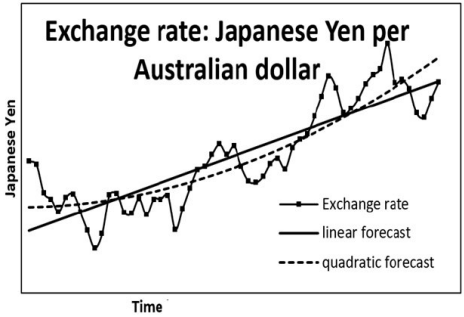


Time Series Model Components



Task 2: Trend Modelling & Forecasting in an Exchange-Rate Time Series

The graph shows monthly values of the exchange rate between the Japanese Yen and the Australian dollar from January 2018 to September 2022. Superimposed on the graph are linear and quadratic curves fitted to this data.



Task 2: Trend Modelling & Forecasting in an Exchange-Rate Time Series

The equations for these linear and quadratic fitted curves are:

Linear fitted curve: $\hat{y} = 58.210 + 0.37t$

Quadratic fitted curve: $\hat{y} = 61.798 + 0.00525t + 0.00629t^2$

Where $\hat{y}$ is the estimated exchange rate in months, and ***t*** is time in months with t=1 corresponding to January 2018.

a) The two equations were used to forecast the exchange rates in the last 3 months of 2022. What is the linear and quadratic forecasts for December 2022?

b) The forecasts for the last three months of 2022 were later compared to the actual values as shown below. Calculate the mean absolute deviation for each forecast, showing your working.

$$\text{MAD} = \frac{\sum_{i=1}^{n} |Y_t - F_t|}{n}$$

Y_t = actual

F_t = forecast

Month	Actual	Linear forecast	Quadratic forecast
Oct-22	79.22	79.68	83.27
Nov-22	80.14	80.05	84.01

Task 2: Trend Modelling & Forecasting in an Exchange-Rate Time Series

The equations for these linear and quadratic fitted curves are:

Linear fitted curve: $\hat{y} = 58.210 + 0.37t$

Quadratic fitted curve: $\hat{y} = 61.798 + 0.00525t + 0.00629t^2$

Where $\hat{y}$ is the estimated exchange rate in months , and ***t*** is time in months with t=1 corresponding to January 2018.

a) The two equations were used to forecast the exchange rates in the last 3 months of 2022. What is the linear and quadratic forecasts for December 2022.

Linear forecast:

t = 60

$\hat{y} = 58.210 + (0.370 * 60)$

$\hat{y} = 80.41$

Quadratic forecast:

t = 60

$\hat{y} = 61.798 + (0.00525 * 60) + (0.00629 * 60^2)$

$\hat{y} = 84.76$

Task 2: Trend Modelling & Forecasting in an Exchange-Rate Time Series

b) The forecasts for the last three months of 2022 were later compared to the actual values as shown below. Calculate the mean absolute deviation for each forecast, showing your working.

Month	Actual	Linear forecast	Quadratic forecast
Oct-22	79.22	79.68	83.27
Nov-22	80.14	80.05	84.01
Dec-22	79.99	80.41	84.76

$$\text{MAD} = \frac{\sum_{i=1}^{n} |Y_t - F_t|}{n}$$

MAD (Linear Forecast)

$$= \frac{|79.22-79.68|+|80.14-80.05|+|79.99-80.41|}{3}$$

$$= \frac{0.46+0.09+0.42}{3} = 0.323$$

MAD (Quadratic Forecast)

$$= \frac{|79.22-83.27|+|80.14-84.01|+|79.99-84.76|}{3}$$

$$= \frac{4.05+3.87+4.767}{3} = 4.229$$

Task 3: Chocolate Bars

A confectionery company tracks quarterly sales of its popular chocolate bars (in thousands). Using historical data from Q1 2017 (t=1), the following linear trend model was used to forecast sales:

$$T=24.369+1.76t$$

Assuming a multiplicative model, the seasonal indices have been calculated.

a. Using trend and seasonality, forecast the sales for the first quarter of 2022.

b. Interpret the seasonal index for the first quarter.

c. Interpret the seasonal index for the fourth quarter.

Q1	0.996
Q2	0.915
Q3	0.934
Q4	1.155

Task 3: Chocolate Bars

a. Using trend and seasonality, forecast the sales for the first quarter of 2022.

$$Ft = T_t * S_t$$

T_{21}=24.369+1.76t = 24.369+1.76*21 = 61.329
S_1 = 0.996
$F_{21} = T_{21} * S_1$ = 61.329 * 0.996 = 61.08 (thousands)

Q1	0.996
Q2	0.915
Q3	0.934
Q4	1.155

b. Interpret the seasonal index for the first quarter.
The sales of chocolate bars in the first quarter is 0.4% below the trend line projection on average.

c. Interpret the seasonal index for the fourth quarter.
The sales of chocolate bars in the fourth quarter is 15.5% above the trend line projection on average.

Task 4: Target Quarterly Sales Revenue

a. Draw a time series plot using the data provided. It shows the quarterly sales revenue for Target from Quarter 1 of 2000 to Q3 of 2016

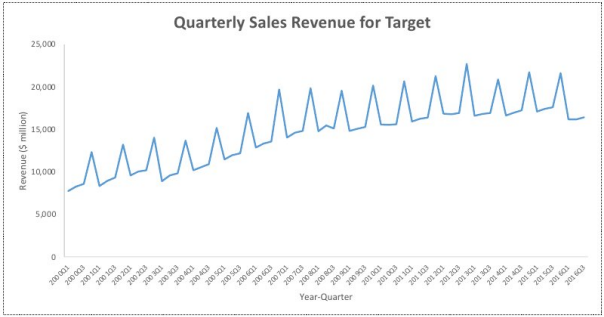
b. Describe the components present in the time series chart.

c. The linear trend model is given by the equation: $\hat{y}_t = b_0 + b_1 t$. Produce the regression results and state the linear trend equation.

d. Interpret the intercept and slope coefficients.

Task 4: Target Quarterly Sales Revenue

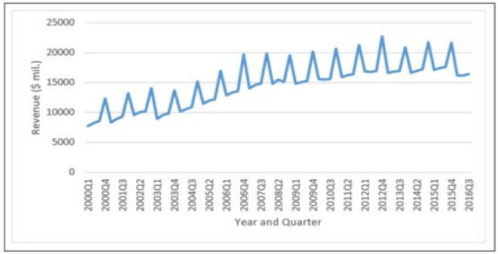
Draw a time series plot using the data provided. It shows the quarterly sales revenue for Target from Quarter 1 of 2000 to Q3 of 2016



Task 4: Target Quarterly Sales Revenue

Describe the components present in the time series chart.

- There is long-term upward trend
- There is a seasonal component present with a very high peak every fourth quarter.
- The shapes of ups and downs vary each year, indicating the presence of an irregular component.



Task 4: Target Quarterly Sales Revenue

The linear trend model is given by the equation:

$$\hat{y}_t = b_0 + b_1 t$$

Produce the regression results and state the linear trend equation.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.80
R Square	0.64
Adjusted R Square	0.64
Standard Error	2275
Observations	67

$$\hat{Y}_t = 9{,}514 + 156t$$

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	9514	562.17	16.92	1.65E-25	8391	10637
Time	156	14.37	10.84	3.27E-16	127	184

Task 4: Target Quarterly Sales Revenue

$$\hat{Y}_t = 9{,}514 + 156t$$

b_0 = 9,514: The predicted **trend value** for sales revenue when Time = 0 is $9,514million.

b_1 = 156: The estimated **average growth** in sales revenue is $156 million per quarter.



- This model is not really adequate.
- There are periods where the predicted and actual values are quite different.
- At least part of the reason is due to seasonality.

Task 5: Forecast, MAD and MSFE

Two forecasting procedures were applied in order to forecast the sales for the four quarters of 2024. The forecasts and the actual sales are given in the following table (Retail sales and forecasts in $million):

Year	Quarter	Sales	Forecast 1	Forecast 2
2024	1	30	31.2	28.6
	2	31	38.0	33.7
	3	40	42.7	36.1
	4	49	54.2	44.4

$$\text{MAD} = \frac{\sum_{t=1}^{n} |Y_t - F_t|}{n}$$

$$\text{MSFE} = \frac{\sum_{t=1}^{n} (Y_t - F_t)^2}{n}$$

a. For each forecasting procedure, calculate the Mean Absolute deviation (MAD) and the Mean Square Forecast Error (MSFE).
b. Based on these measures, which is the best forecasting method? Explain.
c. Let's pretend for a moment that in part a) you found that the MAD measure indicated that Forecast 1 was the most accurate, but the MSFE measure indicated that Forecast 2 was the most accurate. Explain how you would decide which measure of accuracy was the right one to base your decision on.

Task 5: Forecast, MAD and MSFE

For each forecasting procedure, calculate the Mean Absolute deviation (MAD) and the Mean Square Forecast Error (MSFE).

Year	Quarter	Sales	Forecast 1	Forecast 2
2024	1	30	31.2	28.6
	2	31	38.0	33.7
	3	40	42.7	36.1
	4	49	54.2	44.4

$$\text{MAD} = \frac{\sum_{t=1}^{n} |Y_t - F_t|}{n}$$

MAD (Forecast 1)

$$= \frac{|30-31.2|+|31-38|+|40-42.7|+|49-54.2|}{4}$$

$$= 4.025$$

MAD (Forecast 2)

$$= \frac{|30-28.6|+|31-33.7|+|40-36.1|+|49-44.4|}{4}$$

$$= 3.15$$

Task 5: Forecast, MAD and MSFE

For each forecasting procedure, calculate the Mean Absolute deviation (MAD) and the Mean Square Forecast Error (MSFE).

Year	Quarter	Sales	Forecast 1	Forecast 2
2024	1	30	31.2	28.6
	2	31	38.0	33.7
	3	40	42.7	36.1
	4	49	54.2	44.4

$$\text{MSFE} = \frac{\sum_{t=1}^{n}(Y_t - F_t)^2}{n}$$

MSFE (Forecast 1)

$$= \frac{(30-31.2)^2+(31-38)^2(40-42.7)^2+(49-54.2)^2}{4}$$

$$= 21.12$$

MSFE (Forecast 2)

$$= \frac{(30-28.6)^2+(31-33.7)^2+(40-36.1)^2+(49-44.4)^2}{4}$$

$$= 11.41$$

Task 5: Forecast, MAD and MSFE

b. Based on these measures, which is the best forecasting method? Explain.
Both error measures are lower for forecast 2, so based on the 2024 figures, forecast 2 is the more accurate forecasting method.

c. Let's pretend for a moment that in part a) you found that the MAD measure indicated that Forecast 1 was the most accurate, but the MSFE measure indicated that Forecast 2 was the most accurate. Explain how you would decide which measure of accuracy was the right one to base your decision on.

We choose **MSFE** if it is important to avoid (even a few) large errors as it penalizes large errors heavily. Otherwise, use **MAD** as it weights all errors equally and aims for consistent average accuracy.

The choice depends on whether large forecast errors are acceptable in your context.